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TiFLCS-MARP: Client selection and model pricing for federated learning in data markets

Yongjiao Sun [a], Boyang Li [b e f], Kai Yang [a], Xin Bi [c], Xiangning Zhao [d]

[a] School of Computer Science and Engineering, Northeastern University, Shenyang, 110819, China
[b] School of Computer Science and Technology, Beijing Institute of Technology, Beijing, 100081, China
[c] Key Laboratory of Ministry of Education on Safe Mining of Deep Metal Mines, Northeastern University, Shenyang, 110819, China
[d] TianGong University, Tianjin, 300387, China
[e] Tangshan Research Institute, BIT, Tangshan, 063000, China
[f] Hebei Province Key Laboratory of Big Data Science and Intelligent Technology, Tangshan, 063000, China

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Highlights

- Improving the model training performance by optimizing the client selection.
- Proposing a model pricing framework for federated learning in data markets.
- Allocating reasonable payoffs to clients.
- Setting reasonable prices for models based on the optimized model training process.

Abstract

In recent years, the burgeoning data market has witnessed a surge in data exchange, playing a pivotal role in augmenting the predictive and decision-making capabilities of machine learning. Despite these advancements, persistent concerns surrounding data privacy have resulted in stringent limitations on data sharing and trading. Consequently, the data market is undergoing a transformative shift from pricing individual datasets to pricing the models themselves. The challenge of training high-performance machine learning models with restricted data from a single client is substantial. Federated learning has emerged as a popular solution, allowing collaborative model training without the need to transfer client data beyond local environments. However, training federated models within a data market introduces several challenges, including the effective selection of clients for model training and the optimization of utility through model pricing. In response to these challenges, we propose the Tiered Federated Learning Client Selection Algorithm (TiFLCS-MAR), employing a multi-attribute reverse auction approach. Integrated into the federated learning framework, TiFLCS-MAR excels at the comprehensive evaluation of client attributes, employing a tiered strategy to mitigate issues arising from client heterogeneity. Additionally, we introduce the TiFLCS-MAR Pricing Framework (TiFLCS-MARP), leveraging Nash equilibrium principles to maximize profitability for both clients and servers. Our framework accommodates the heterogeneity of diverse clients, efficiently selecting suitable candidates from a large pool, thereby boosting training efficiency and curbing model pricing costs. Empirical evidence showcases the efficacy of federated training with TiFLCS-MAR, demonstrating nearly double the convergence speed and a 5-10 percentage point improvement in accuracy across real and synthetic datasets. Furthermore, when compared to three baseline algorithms, TiFLCS-MARP substantially increases central server revenue by factors of 1.99, 27.05, and 1.78, highlighting its superior performance in the data market context.

Introduction

With the rapid development of information technology, various fields have generated rich data, and big data has now become a core resource in various fields. To make profits and render data valuable, the information can be sold to other organizations. In this context, big data trading platforms and data marketplaces have emerged (Niu et al., 2022). However, datasets are usually extremely expensive because financial and material resources are usually required to collect, integrate, and clean the data before they can be officially used. This implies that potential buyers with budgets may not be able to afford expensive datasets, resulting in data sellers operating in an inefficient marketplace. Meanwhile, not all data are available for trading owing to privacy concerns.

The above-mentioned problems render it more necessary and urgent for the data market to transition; thus, a new pricing model is proposed. In this model, the ML model instances instead of data are directly priced (Chen et al., 2019). However, there is still a prevalent problem: in many scenarios, data are accompanied by privacy constraints, and even within the same company, data from various departments are generally separated from each other. Each department's data do not (or cannot easily) interact with those of other departments, resulting in silos. Therefore, it is difficult for individual data owners to train appropriate models with their local data.

Federated learning can alleviate the above-mentioned data separation problem. There are three major components of federated learning: data source, federated learning system, and users. In a federated learning system, individual data source parties perform data preprocessing while keeping the data localized (Yang et al., 2019), jointly build the model, and feed the output to users. In the data markets, data owners are *Clients* who provide data to train the federated learning model, the central server are *Brokers* that coordinate the training process and price the model, model buyers are consumers who pay for the model. In this work, our goal is to address the federated learning process and model pricing in the data markets. However, given the specificity of the federated model training process, implementing pricing for federated models presents the following challenges:

- First, the number of clients that can participate in federated learning is large, and if all clients participate in federated learning, not only is the model training negatively impacted but also the model training cost increases.

- Second, due to the differences in computing and communication capabilities, as well as the differences in data volume and content, the clients are heterogeneous.
- Furthermore, the training cost of different clients is not consistent, and the participation of clients with high training cost lead to higher model price.

Therefore, we improve the model training performance by optimizing federated learning client selection and proposes a model pricing framework for federated learning in data markets. As shown in Fig. 1, the framework allocates reasonable payoffs to clients and sets reasonable prices for models based on the optimized model training process. Step (1)–(4) represent the training process, Step (4) represents the server's process of allocating payoffs to participating clients, Step (5) represents the model pricing process, Step (6) represents the server's process of publishing model instances and their prices to the market, Step (7) represents the model buyers' purchase of target models and payment of prices according to their budgets, and Step (8) represents the final model transactions. The contributions and originality are as following:

- We propose a novelty algorithm, TiFLCS-MAR, to address the heterogeneity of clients in federated learning, enhancing the training effectiveness of federated learning.
- We introduce a federated learning pricing framework for data markets, TiFLCS-MARP, which, by considering client heterogeneity, establishes a rational pricing mechanism for federated learning models.
- Through experiments conducted on real and synthetic datasets, the results demonstrate that our approach achieves superior training outcomes while simultaneously enhancing profits in the data market.

This paper is organized as follows. Section 2 reviews the existing researches. Section 3 defines the optimization problem of federated learning client selection and the utility maximization problem in the pricing framework. Section 4 describes the optimized client selection algorithm and the corresponding federated learning process. Section 5 proposes a novel model pricing framework. Section 6 verifies the necessity of this research through experiments and validates the effectiveness and robustness of the proposed algorithm through experiments on several datasets and proves the effectiveness of the proposed pricing framework via numerical analysis. Section 7 provides a summary of the entire study and suggests a few potential future directions of research.

Section snippets

Related works

This section summarizes the existing research related to the current study in three aspects: (1) federated learning training objectives, (2) federated learning client selection, and (3) data market pricing.

Problem definitions

This section presents the notations and related definitions for the federated learning client selection and model pricing.

Definition 1 Client…

Let $\{client_1, \quad \cdots, \ client_N\}$ be a set of data owners, where each data owner has its own local data source $\{D_1, \quad \cdots, \quad D_N\}$, and such data owners are defined as clients.

Clients who can participate in federated training can be mobile devices, enterprises, or individuals. Each client is willing to participate in federated training with local datasets under the premise of reasonable payoff,

Solving straggler problem

In this study, we divide the clients into different levels according to the response latency for the straggler problem mentioned above based on the tiered concept presented previously (Chai et al., 2020), and the clients are selected in the same layer to participate in model training to solve the straggler problem. The tiered process is to complete a small federated learning task, and this study assumes that the hierarchies of clients are uniformly distributed. As a first step, the training

TiFLCS-MARP framework

In this part, we construct a new pricing algorithm (MARP) based on the TiFLCS that maximizes the revenue of the central server while providing each client with a Nash equilibrium strategy that satisfies individual rationality and incentive compatibility. The main tasks include payoffs to the participating clients and pricing of the final federated model. The pricing is for the model training process, not for the end result.

Experiments

In this study, two experiments were conducted to verify whether the TiFLCS-MAR algorithm can afford time savings and improved final convergence accuracy to the federated learning training process and to verify if the pricing method in the proposed pricing framework can effectively increase the server revenue. Next, we discuss the experiments in detail.

Conclusion

Our research addresses the evolving landscape of data markets, where concerns about data privacy have necessitated a shift from data pricing to model pricing. We introduced TiFLCS-MAR, a multi-attribute reverse auction-based tiered federated-learning client-selection algorithm, to enable efficient and effective client selection. TiFLCS-MAR not only streamlines the training of high-performance ML models without exposing raw data but also tackles the challenge of client heterogeneity, ensuring

CRediT authorship contribution statement

Yongjiao Sun: Conceptualization, Methodology, Project administration. **Boyang Li:** Formal analysis, Writing – original draft. **Kai Yang:** Software. **Xin Bi:** Validation, Resources, Writing – review & editing. **Xiangning Zhao:** Validation, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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